

Nine-Axis IMU-based Extended inertial odometry neural network

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ARTICLE INFO

Keywords:

Extended inertial odometry neural network
Inertial measurement unit
Drift
Pose-TuningNet

ABSTRACT

With the development of mobile devices, such as smartphones, research on fast and accurate trajectory tracking is being actively conducted. This research requires a continuous integration of the acceleration and angular velocity data obtained from the low-cost microelectromechanical system-based inertial measurement unit (IMU) installed in a device to track the user's trajectory. During this process, drift occurs over time due to the bias and intrinsic error of the IMU sensor. Hence, the 6-Axis IMU-based inertial odometry neural network (IONet) using deep learning, which is designed as a framework for velocity estimation, is used to reduce drift by dividing the acceleration data into independent windows. However, drift still occurs in estimating a pose containing both a position and an orientation because the integration of pose changes is also required. In this study, we proposed the Extended IONet that combines a 9-Axis IONet and Pose-TuningNet to improve the accuracy of trajectory tracking by compensating for the drift problem of the 6-Axis IONet. The proposed 9-Axis IONet uses the gravitational acceleration and geomagnetic data of the IMU in addition to the input structure of the existing 6-Axis IONet; thus, the estimation accuracy of pose changes improves by reducing the data dependence on the original input of the 6-Axis IONet. The proposed Pose-TuningNet is an auxiliary network that is capable of estimating pose changes more precisely using the higher-dimensional inclination-angle information obtained from the IMU to focus on the noise model of the IMU. Experiments were conducted using the Oxford Inertial Odometry Dataset, which is public dataset for deep learning based inertial navigation research to verify the performance of the proposed neural network. Compared with the existing 6-Axis IONet, the Extended IONet achieved superior performance in five out of seven cases, and its overall 39.8% RMSE improvement demonstrated its excellent performance. Additionally, the results showed that Pose-TuningNet improved the position estimation performance by correcting the drift problem in the 9-Axis IONet.

1. Introduction

The main challenge encountered in odometry technology is designing devices and algorithms that can accurately track trajectories regardless of movement. These essential trajectory tracking technologies are used in a wide range of applications, such as unmanned aerial vehicles (Altan & Hacıoğlu, 2020), robots (Belter & Nowicki, 2019), biomedical engineering (Wang et al., 2019), sports training (Ziha-jehzadeh et al., 2015), and augmented reality (Marchand et al., 2015). The wide adoption of the trajectory tracking technologies is facilitated by the rapid development of small sensors and mobile computing technologies. Moreover, several studies are actively underway and mainly based on visual odometry (VO), which uses a camera sensor for

trajectory tracking by estimating the exact odometry of a moving object (Scaramuzza & Fraundorfer, 2011). This method has the advantage of low drift error compared with other position recognition sensors; however, its accuracy may be degraded by camera shake, moving objects, and lighting changes. To compensate for the accuracy degradation, the visual inertial odometry (VIO) in which the inertial measurement unit (IMU) is integrated is employed to stabilize the trajectory tracking (Forster et al., 2016; Li & Mourikis, 2012; Chan et al., 2017).

Several studies have been conducted on Inertial Odometry (IO) technology involving only an IMU sensor without a camera, unlike VO or VIO, which involves a large number of computations owing to feature extraction and the mapping of hundreds of thousands of pixels. The primary problem in IO is the simultaneous estimation of the intrinsic

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<https://doi.org/10.1016/j.eswa.2021.115075>

Received 18 September 2020; Received in revised form 18 March 2021; Accepted 17 April 2021

Available online 21 April 2021

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error, bias, and gravity direction from acceleration readings to accurately calculating the linear acceleration caused by sensor motion only. Therefore, a simple approach based on the double integration process of acceleration can lead to severe drift errors.

To address this problem, strapdown inertial navigation system (SINS) (Savage, 1998; Titterton et al., 2004; Wu et al., 2005) and pedestrian dead reckoning (PDR) (Nilsson et al., 2013; 2014; Lu et al., 2019) have been proposed. SINS estimates the velocity by integrating the acceleration without the gravity component after estimating the posture from the acceleration and gyro data measured by an IMU. Several difficulties have been encountered in using this system because sensor bias and intrinsic errors cause drift during acceleration integration, infinitely diverging the estimated position over time. Meanwhile, as PDR estimates the trajectory by using the IMU attached to the body (e.g., the foot) to determine the position of the pedestrian, it can compensate for drift by analyzing periodic behavior patterns such as gait. However, because pedestrians also have irregular behavior patterns that may occur in their daily life, the analysis and utilization of these complex and diverse behavior patterns have limitations. Additionally, when the IMU sensor is not fixed (e.g., placed in a bag or a pocket or held in the hand), the orientation of the device may change, resulting in errors, thus degrading the trajectory tracking accuracy.

Recently, deep-learning-based inertial odometry neural networks (IONets) can address the drift problem of inertial localization techniques using supervised learning to directly estimate the spatial displacement of the device (Chen et al., 2018; Abolfazli Esfahani et al., 2020; Silva do Monte Lima et al., 2019). Since the deep-learning technology has yielded high accuracy in classification tasks for various types of data, such as images, text, or sound, it is used in various industries ranging from autonomous driving (Li et al., 2019) and medical devices (Alhussein et al., 2019) to finance (Karasu et al., 2020; Altan et al., 2019). Hence, IONets implicitly reduce the intrinsic error of the sensors by modeling the complex nonlinear relationship between the raw sequence data of the IMU and the pose containing both a position and an orientation. Additionally, by designing continuous sequence data as a window, the drift that occurs in the integration acceleration process was reduced.

However, because drift still occurs during the process of integrating velocity and orientation changes, there is a need for research on this topic. Moreover, most IONets in the literature use only the linear acceleration and gyro sensor data, excluding the gravitational acceleration and geomagnetic data that can be obtained from a 9-Axis IMU. For high-precision trajectory tracking using only a single IMU, the gravitational acceleration and geomagnetic data in IONets should be utilized.

Therefore, in this study, we proposed the extended inertial odometry neural network (Extended IONet) that combines a 9-Axis IONet and Pose-TuningNet to improve the accuracy of trajectory tracking by compensating for the drift problem of the 6-Axis IONet. The proposed 9-Axis IONet uses the gravitational acceleration and geomagnetic data of the IMU in addition to the input structure of the existing 6-Axis IONet; thus, the estimation accuracy of pose changes improves by reducing the data dependence on the original input of the 6-Axis IONet. The proposed Pose-TuningNet is an auxiliary network that is capable of estimating pose changes more precisely using the higher-dimensional inclination-angle information obtained from the IMU to focus on the noise model of the IMU. The proposed neural network can improve the estimation performance for pose changes by sharing features related to orientation with the IONet through Pose-TuningNet. The advantage of this method is that it reduces drift by incorporating gravitational acceleration and geomagnetic data, unlike existing IONets. The model can be generalized by continuously learning the orientation through the convolution neural network (CNN) connected to the gravitational acceleration data using Pose-TuningNet.

The major contributions of this work are summarized as follows:

- We propose a novel IONet that infers pose changes from the IMU raw data by efficiently learning from the gravitational acceleration

(including inclination angle information) and geomagnetic (including azimuth information) data.

- The proposed Pose-TuningNet is a new learning framework that can improve the estimation performance for pose changes by calculating gravitational acceleration into inclination information.
- Our proposed method is easy to apply in various existing IONets by incorporating the gravitational acceleration and geomagnetic data that have been previously excluded.

The rest of this paper is organized as follows. Section 2 provides an overview of related work, and Section 3 presents the basis for including gravitational acceleration and geomagnetic data as input to the IONet as well as a method for representing 6-DOF relative poses. We describe the proposed Extended IONet to reduce drift in Section 4 and discuss how to design the experiments and details in Section 5. The performance evaluation and comparison results are presented in Section 6, and we provide the conclusions and scope for future work in Section 7.

2. Related work

2.1. Data-Driven inertial method for inertial navigation

The data-driven method, which has been used for various types of inertial tracking tasks, is an approach that can determine the connections between system state variables without knowledge of the physical behavior. Inertial-based gesture recognition was achieved through a probabilistic model using a 9-Axis IMU (Parate et al., 2014). Mannini and Sabatini (2014) estimated human walking speeds with the hidden Markov model-based strapdown integration method. Owing to the advancements in machine learning techniques, IMUs have also been used for analyzing walking and poses (Mannini & Sabatini, 2010; Valtazanos et al., 2013; Yuwono et al., 2014) as well as gait and posture (Nilsson et al., 2013; 2014). Additionally, the most used gait period is the single support mid-stance phase during which the speed of the inertial measurement unit (IMU) mounted on the foot equals zero. Hannink et al. (2018) presented a method to train a CNN to predict human stride length from Foot-mounted IMU. To adaptively update the zero-velocity threshold, Wagstaff et al. (2020) proposed learning-based classifiers model that learns the relationship between raw inertial data and latent variables such as the wearer's walking motion type (i.e., walking, running, stair-climbing) or the velocity state of the IMU. Kone et al. (2020) designed a zero-velocity detection method using histogram-based gradient boosting and random forest to reduce computational cost compared with deep learning-based detectors and for various types of motion. Unlike other methods that unconditionally require a foot-mounted sensor, Yan et al. (2018) categorized phone placement into four (i.e., pocket, bag, hand, body) and regressed velocity vector from linear acceleration and angular velocities. These studies estimated human motion at a fixed IMU position and by restricting some of the IMU poses. Unlike in previous studies, our proposed model can learn motion expression from data in an end-to-end manner and reconstruct accurate three-dimensional (3D) trajectories using a network that directly estimates poses.

2.2. Inertial odometry neural network

IONets are end-to-end learning framework for inertial odometry to track pose of an object in a navigation frame given an initial pose and measurements from accelerometers and gyroscopes. Additionally, They don't have any assumptions about sensor placement and user motion; hence, it can resolve the limitations of existing methods for inertial indoor navigation. Chen et al. (2018) proposed a long short-term memory (LSTM)-based deep learning framework that estimates the velocity by dividing the acceleration data into independent windows. by designing continuous sequence data as a window, the drift that occurs in the integration acceleration process was reduced. This method obtained

good results in human pose estimation; however, the architecture is not sufficiently robust to handle the noise and frequency changes of the IMU as it does not model the uncertainty and noise. Hence, Abolfazli Esfahani et al. (2020) improved the estimation of speed and orientation by using three LSTMs (i.e., acceleration and gyro data, and sampling time), and features were extracted according to each input. Meanwhile, Lima, Uchiyama, and Taniguchi (2019) solved the problem encountered in six-degree-of-freedom (6-DOF) relative pose representation. For example, unlike vehicles, humans can move backward, sideways, or forward; hence, representing humans using the 6-DOF relative pose representation based on the spherical coordinate system is a difficult task. Hence, a deep learning model was designed based on a translation vector and unit quaternion to address this problem. Additionally, Chen et al. (2020) designed a lightweight framework as a mobile platform to reduce the numerous computations of deep learning models.

This system estimates poses by integrating the amount of pose change, which is the output of the network inputted with only free acceleration and gyro, resulting in continuous drift. Consequently, the number of orientation changes is difficult to estimate accurately; thus, errors in the moving direction accumulate. The proposed Extended IONet reduces drift by adding a network that directly estimates orientation using measurements from gravitational acceleration and geomagnetism, enabling it to estimate accurate and robust trajectories.

3. Background

3.1. Orientation measurement based on gravitational acceleration

Gravitational acceleration is the acceleration of an object caused by the gravity of the earth, and it acts in a direction perpendicular to the earth surface. Because the earth is not a perfect sphere, gravitational acceleration varies per region. However, it has an average value of $9.8m/s^2$.

$$\vec{g} = \vec{A}_x + \vec{A}_y + \vec{A}_z \quad (1)$$

Gravitational acceleration $\vec{g}$ is a composite of the vectors of acceleration $\vec{A}_x$, $\vec{A}_y$, and $\vec{A}_z$, as shown in (1), and varies depending on the orientation of an IMU. Fig. 1 illustrates the force that the IMU receives when inclined by the roll angle θ and the pitch angle ϕ , which is the rotation angle on the x-axis and the y-axis. Because the x-axis rotates, θ

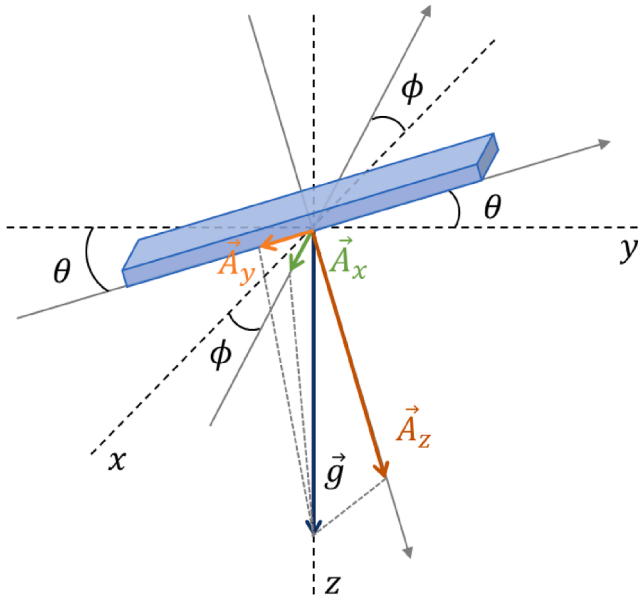


Fig. 1. Gravity accelerometer.

can be calculated using the sizes of $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$, as in (2). The pitch angle ϕ , which is the rotation angle on the y-axis, can be expressed through $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$, as observed in (3). The linear acceleration used in conventional IONet is calculated by excluding the gravitational acceleration from the acceleration. At this time, since the ratio of gravitational acceleration can be changed by intrinsic error or movement, using only linear acceleration excluding gravitational acceleration in IONet is a factor that can degrade position estimation performance.

$$\theta = \tan^{-1} \frac{\vec{A}_y}{\sqrt{\vec{A}_z^2 + \vec{A}_x^2}} \quad (2)$$

$$\phi = \tan^{-1} \frac{\vec{A}_x}{\sqrt{\vec{A}_z^2 + \vec{A}_y^2}} \quad (3)$$

3.2. Azimuth measurement based on geomagnetism

The magnetic vector of the earth has seven components, as shown in Fig. 2. Here, X refers to the geographic north (true north), that is, the north end of the rotation axis of the earth; Y to the geographic west; Z to the vertical magnetic force toward the nadir; F to the magnetic force; H to the horizontal magnetic force (in the magnetic north direction); I to the angle between the magnetic force and horizontal plane (inclination); and D to the angle between true north and magnetic north (declination).

Equation (4) represents the magnetic force F of the earth, and (5) represents the horizontal magnetic force H.

$$F = \sqrt{H_d^2 + R_d^2 + N_d^2} \quad (4)$$

$$H = \sqrt{H_d^2 + R_d^2} \quad (5)$$

Here, H_d , R_d , and N_d refer to the heading, right, and nadir directions, respectively. To calculate the orientation angle, the horizontal coordinate system transformation according to the current orientation is essential and shown in (6).

$$\begin{pmatrix} X_H \\ Y_H \\ Z_H \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sin\phi & 0 \\ 0 & 0 & \cos\phi \end{pmatrix} \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} \quad (6)$$

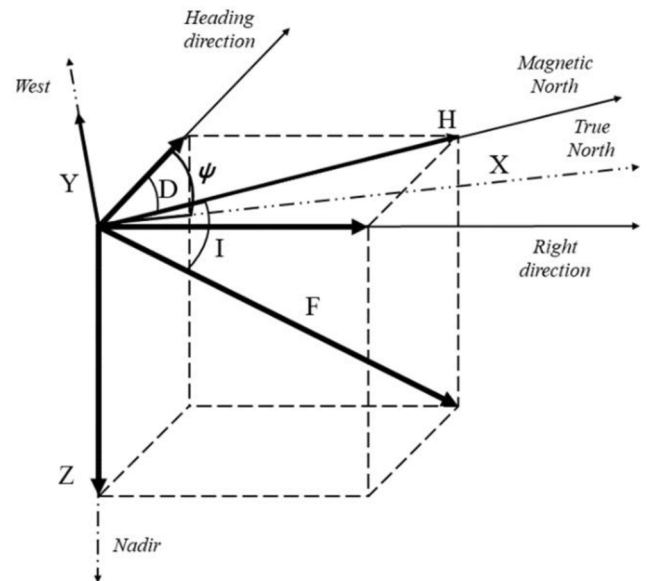


Fig. 2. Components of magnetic vectors.

where θ and ϕ are the roll and pitch angles, respectively; m_x , m_y , and m_z are measured magnetic fields; and X_H , Y_H , and Z_H are magnetic vector values where the orientation is compensated. These values can be used to calculate the azimuth ψ , as shown in (7) (Seong et al., 2019).

$$\psi = \tan^{-1} \left(\frac{m_y \cos \theta - m_z \sin \phi}{m_x \cos \theta + m_y \sin \phi \sin \theta + m_z \cos \phi \sin \theta} \right) + D. \quad (7)$$

3.3. 6-DOF relative pose representation

In this paper, among the various 6-DOF pose representations, we used the 3D translation vector Δp , unit quaternion Δq , and previous orientation q_{t-1} to represent pose changes, as in the paper of Silva do Monte Lima et al. (2019). This approach involves calculating the current position and orientation by adding the amount of change between unit times to the previous position and orientation, as shown in (8).

$$\begin{cases} p_t = p_{t-1} + R(q_{t-1})\Delta p \\ q_t = q_{t-1} \otimes \Delta q \end{cases} \quad (8)$$

where p_t is the current position calculated by adding the amount of pose change between unit times (i.e., the product of the rotation matrix $R(q)$ and Δp) to the previous position p_{t-1} ; the current orientation q_t is the sum of the poses through the Hamilton product $\otimes$ between the previous orientation q_{t-1} and Δq (Parcollet et al., 2020). The quaternion estimated from the neural network is normalized to maintain the unit quaternion form representing the pose, as shown in (9).

$$u(q) = \frac{q}{\|q\|} = \frac{w}{q} + i \cdot \frac{x}{d} + j \cdot \frac{y}{d} + k \cdot \frac{z}{d} \quad (9)$$

Where i, j, k are unit vectors orthogonal each other and $d = \|q\| = \sqrt{w^2 + x^2 + y^2 + z^2}$ is the norm. The advantage of this system is that it can represent the movement path, such as backward or sideways movement, without changing the direction.

4. Extended inertial odometry neural network

Our proposed Extended IONet aims to improve the trajectory-tracking performance using a network that can efficiently utilize the data acquired from the 9-Axis IMU. Fig. 3 illustrates the overall framework of the method proposed herein. The proposed method is divided into an encoding step, wherein each feature of the inertial and geomagnetic output from the 9-Axis IMU is encoded, and a decoding step, wherein all the encoded features are integrated and a pose change is generated.

4.1. Nine-Axis IONet

To calculate the change in orientation accurately, information about the previous orientation is important; hence, the performance of IONet can be improved by learning related orientation information together. As described in Section 3, gravitational acceleration is used to calculate the roll angle θ and pitch angle ϕ . Based on these quantities, the geomagnetic data are then used to calculate the yaw angle, ψ . This indicates that gravitational acceleration and geomagnetism are directly related to the orientation. In addition, since geomagnetism can be converted to azimuth angle only when information about the inclination angle is provided as shown in Equation (7), it is necessary to design a learning network by integrating the gravitational acceleration related to the inclination angle. Therefore, we proposed the 9-Axis IONet, which inputs the 9-Axis-IMU-based-linear acceleration, gyroscope, gravitational acceleration, and geomagnetic data to physically consider the orientation.

Fig. 4 illustrates the structure of the 9-Axis IONet. the input data of the 9-Axis IONet comprise linear acceleration a , angular velocity w , gravitational acceleration g , and geomagnetic value m . Similar to the study of Silva do Monte Lima et al. (2019), the time-series data inputted

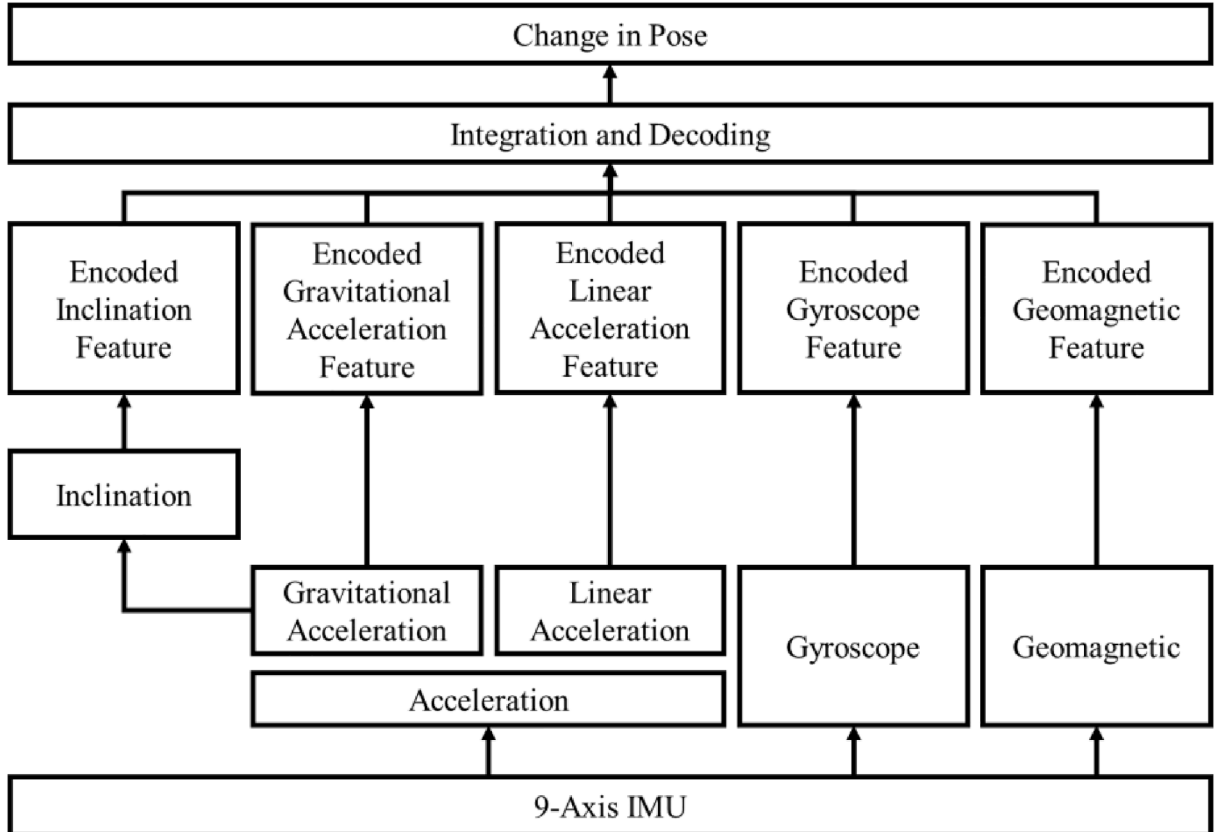


Fig. 3. The overall framework of method proposed Extended IONet.

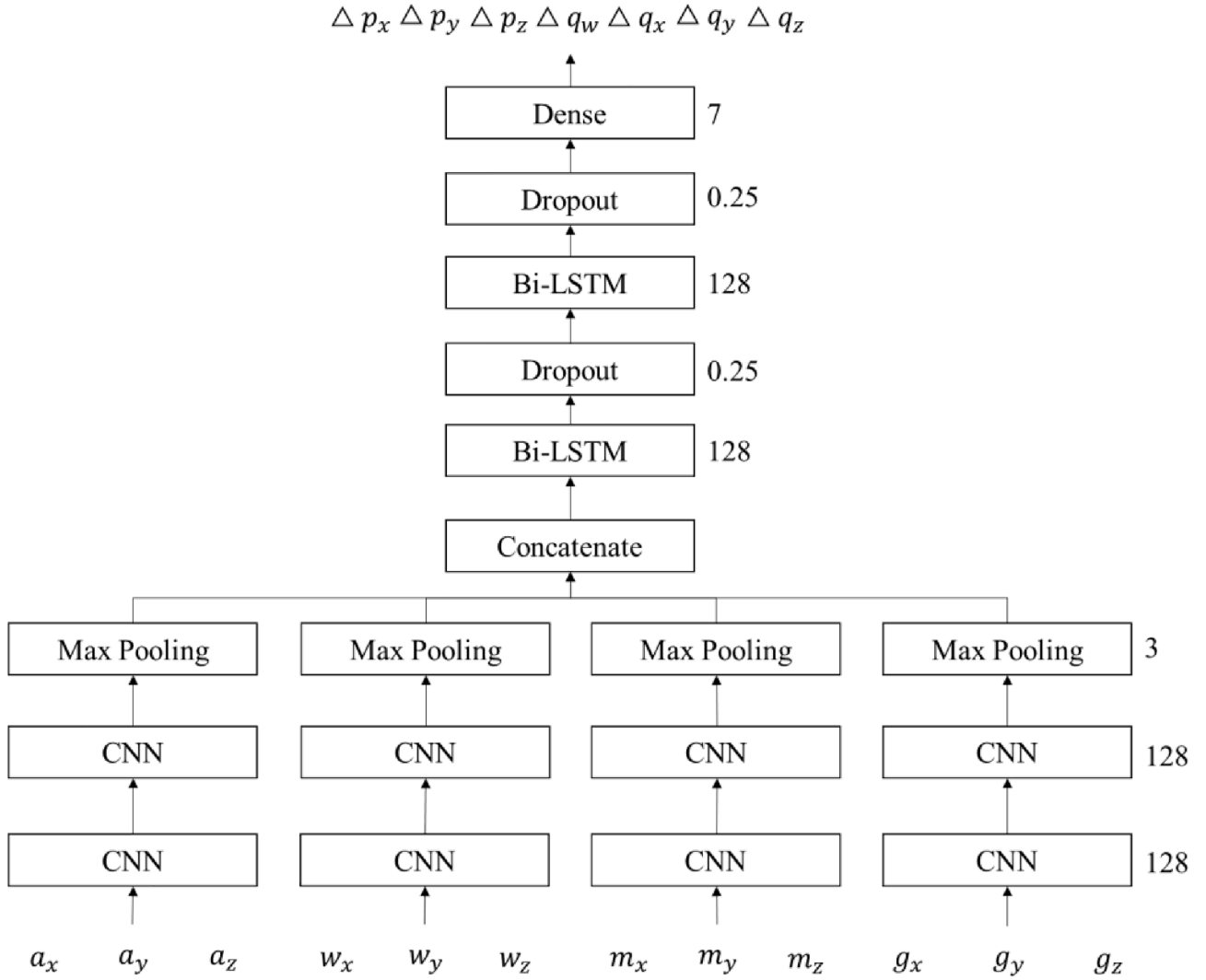


Fig. 4. Structure of 9-Axis IONet.

to the proposed network in the 1D form are generated as one set of 200 frames for every 10 strides of continuous IMU data; Δp and Δq are calculated as the difference between the 95th and 105th data points in one set of data. Although this approach has a slight delay compared to the input time, it obtains slightly improved performance by considering the input of the time series in the two directions of the previous frame and the subsequent frame.

The structure of the proposed 9-Axis IONet contains a CNN layer, a bi-directional long short term memory (Bi-LSTM) layer, and a fully connected layer (Dense). Because the input data are of different types, each two-layer one-dimensional CNN layer is used to encode meaningful feature representations to extract the important information from the 9-Axis IMU. In the next step, a 3×3 max pooling is performed to reduce the computational complexity and extract a strong feature. Subsequently, data are integrated into the Concatenate layer, which combines the feature maps of a , w , g , m . Finally, Δp and Δq are estimated through the two Bi-LSTM layer and the Dense layer. Therefore, as the 9-Axis IONet can estimate the changes in orientation more accurately than the 6-Axis IONet can, the performance of position-change estimation is improved.

4.2. Pose-TuningNet

In the process of estimating a pose using IONet from raw data of IMU,

reliability may be degraded because the intermediate process cannot be known. Therefore, we propose an Extended IONet that combines 9-Axis IONet and pose-tuningNet to learn directly about sensor noise by inputting information that can be calculated easily physically such as orientation. Fig. 5 introduces the architecture of the proposed model namely pose-tuningNet for improving the performance of pose prediction. Pose-TuningNet expands the gravitational acceleration to raw data as well as data converted to orientation, and each input data generates a feature map using CNN layer and Maxpooling. The orientation roll angle θ and the pitch angle ϕ are calculated using equations (2) and (3) for the gravitational acceleration, respectively. Finally, in order to match the 3-dimensional representation of the IONet's orientation-related output data, the data output at the euler angle is converted into quaternion q .

$$q(\theta, \phi, \psi) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\phi}{2}\right)\cos\left(\frac{\psi}{2}\right) + \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\phi}{2}\right)\sin\left(\frac{\psi}{2}\right) \\ -\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\phi}{2}\right)\sin\left(\frac{\psi}{2}\right) + \cos\left(\frac{\phi}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\phi}{2}\right) + \sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\phi}{2}\right)\sin\left(\frac{\psi}{2}\right) \\ \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\phi}{2}\right)\sin\left(\frac{\psi}{2}\right) - \sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\phi}{2}\right) \end{bmatrix} \quad (9)$$

Here, gravitational acceleration is related to roll and pitch, so yaw ψ is

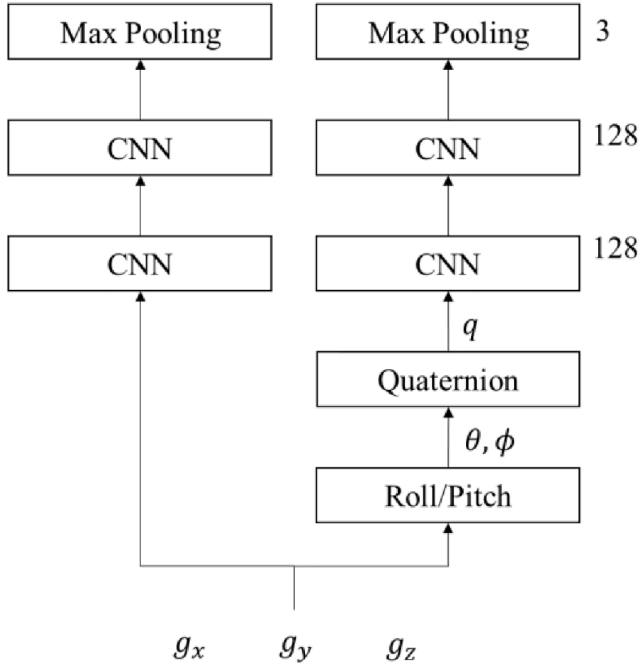


Fig. 5. Structure of Pose-TuningNet.

0. By adding a task that can be easily converted physically through this process, only the orientation estimation error can be considered, so the attitude estimation performance of IONet can be improved.

4.3. Loss function

Because the output of the proposed model has a different scale, the loss for Δp and Δq should be treated as separate tasks. Therefore, we aimed to train each output equally with the total loss function L , using multitask learning as described in Kendall et al. (2018).

$$L_{total} = \sum_{i=1}^n e^{-\log \sigma_i^2} L_i + \log \sigma_i^2 \quad (10)$$

where σ_i^2 and L_i are the variance and loss function of the i th task, respectively. As the loss avoids division by zero, predicting the log variance is more numerically stable than regressing the variance (Kendall et al., 2018). Three loss functions are used in the calculation of L .

$$[L_1, L_2] = [L_{DPMSE}, L_{DQME}] \quad (11)$$

Here, L_{DPMSE} is the delta position mean absolute error and L_{DQME} is the delta quaternion mean error:

$$L_{DPMSE} = \|\Delta \hat{p} - \Delta p\|_1 \quad (12)$$

$$L_{DQME} = 2 * \|\text{image}(\Delta \hat{q} \otimes \Delta q^*)\|_1 \quad (13)$$

L_{DPMSE} and L_{DQME} are the loss functions for the velocity and orientation changes in the IONet, respectively, and the model is trained for the geometric change between the ground truth ($\Delta p, \Delta q$) and predicted pose ($\Delta \hat{p}, \Delta \hat{q}$). As Δp is the differential value of the trajectory, and Δq is the value representing the amount of the orientation change in the IMU, the absolute value of the 3D pose graph simultaneous localization and mapping error is used as the loss function (Silva do Monte Lima et al., 2019). Here, Δq^* is the conjugate complex number of Δq , $\text{image}(q)$ returns a complex number as a real number. The $\Delta \hat{q}$ and $\hat{q}$ estimated from the neural network are in the form of a unit quaternion with the size of 1.

5. Experiments

5.1. Dataset

The ground truth was created using a Vicon motion capture system on the Oxford Inertial Odometry Dataset (OxIOD) (Chen et al., 2018). The data were measured at four attachment locations using an iPhone 7 Plus, which has a low-cost IMU and geomagnetic sensor. In the experiments, handheld data with the largest 3D pose change were used; the data comprised 24 data sequences measured in different conditions at a sampling rate of 100 Hz. As presented in Table 1, we divided the data into 17 training and 7 testing data sequences to compare the performance of the proposed method in the same experimental environment as Silva do Monte Lima et al. (2019), and we conducted the experiment after discarding the initial 12 s and last 3 s of each data sequence because of measurement noise in the ground truth. One input data sequence contained 200 frames, a stride of 10 frames was added between data sequences, and there was a total of 55,003 input training samples. We performed qualitative and quantitative model evaluations after obtaining the output data using the initial 20 s of the testing data sequences.

5.2. Training

Keras 2.3.1 and TensorFlow 1.13.1 were used in the training framework, and an Nvidia RTX Titan was used as the GPU. During training, Adam optimizer with a learning rate of 0.0001 was used, the validation data was set to 10% of the training data, the network was trained 500 times, and the batch size was 32.

5.3. Evaluations

To evaluate the performance of the proposed 9-Axis IONet and Extended IONet, we compare with the 6-Axis IONet (Silva do Monte Lima et al., 2019) that is superior to state-of-the-art inertial odometry techniques such as IONet (Chen et al., 2018), VINet (Clark et al., 2017), AbolDeepIO (Abolfazli Esfahani et al., 2020). To analyze both trajectory and orientation results, we use a model trained directly in the same experimental environment for the performance evaluation of the 6-Axis IONet, without using the results proven by Silva do Monte Lima et al. (2019). The 6-Axis IONet, 9-Axis IONet, and Extended IONet were trained using the loss function for Δp and Δq . For the qualitative evaluation, we used two datasets, namely “handheld/data1/seq2” and “handheld/data4/seq3,” obtained from visually circular trajectories and more complex trajectories. Each plot shows the aligned ground truth (red curves) and predicted trajectories (blue curves). The trajectories can be viewed from both the top and the sides; thus, the 6-DOF pose can

Table 1
Training and test splits of OxIOD handheld sequences.

Training Data	Testing Data
data1/seq1	data1/seq2
data1/seq3	data1/seq5
data1/seq4	data1/seq6
data1/seq7	data3/seq1
data2/seq1	data4/seq1
data2/seq2	data4/seq3
data2/seq3	data5/seq1
data3/seq2	
data3/seq3	
data3/seq4	
data3/seq5	
data4/seq2	
data4/seq4	
data4/seq5	
data5/seq2	
data5/seq3	
data5/seq4	

be estimated. We also performed the quantitative evaluation of each model using the position and the orientation errors between each model output and the ground truth by inputting seven test data sequences.

6. Results and discussion

6.1. Qualitative evaluation

Fig. 6 presents the trajectory-estimation results processed by 6-Axis

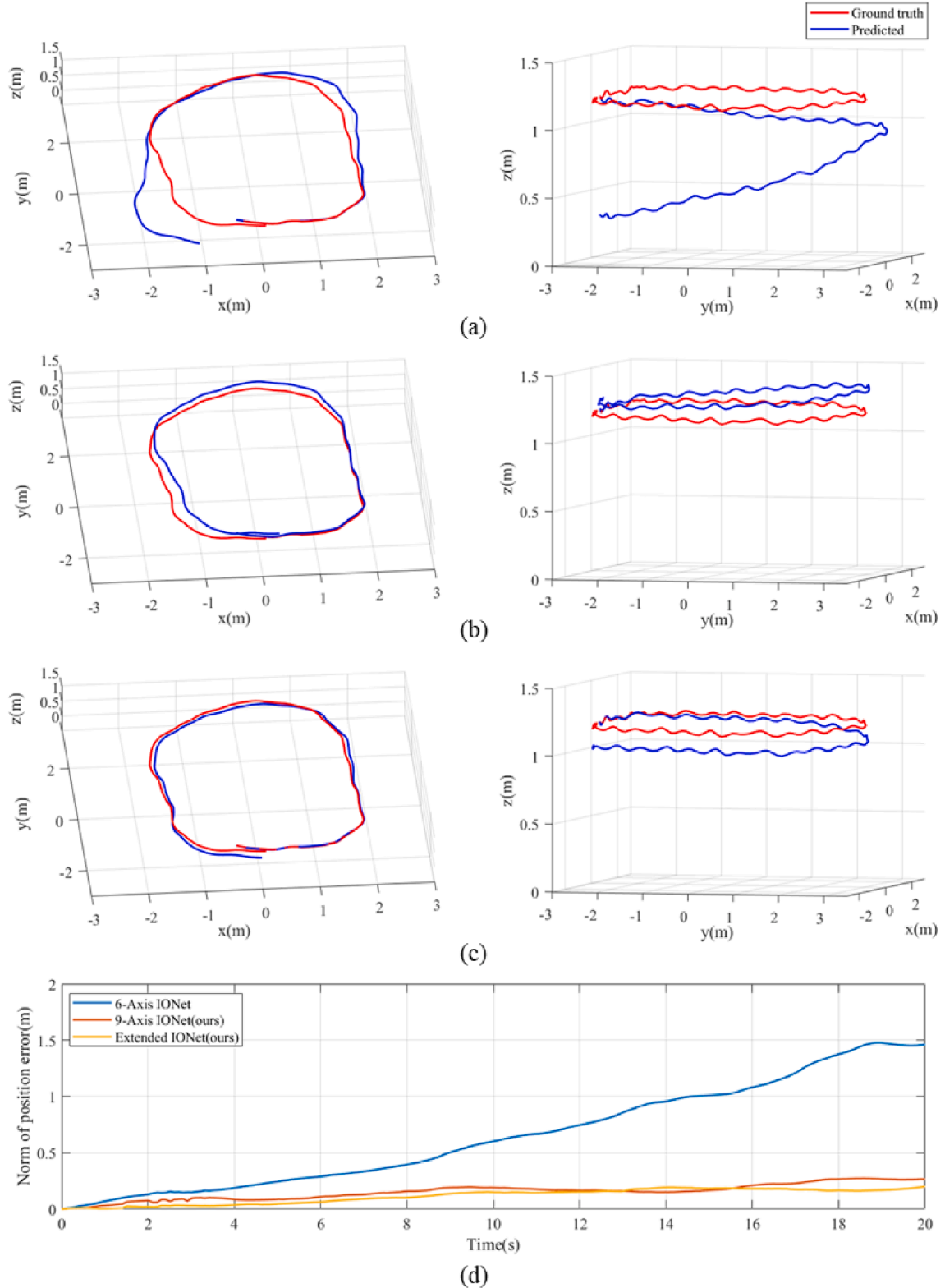


Fig. 6. Top (left column) and side (right column) views of ground truth (blue) and predicted (orange) trajectories for an excerpt of the "handheld/data1/seq2" dataset using (a) 6-Axis IONet, (b) 9-Axis IONet, (c) Extended IONet, and (d) the position error of the trajectories.

IONet (a), 9-Axis IONet (b), and Extended IONet (c) using the “handheld/data1/seq2” dataset. Fig. 6(d) shows the position error of the trajectories created by the 6-Axis IONet (blue line), 9-Axis IONet (orange line), and Extended IONet (yellow line). The analyzed position error corresponds to the norm of position errors, calculated as $\Delta e_p = \|\hat{p} - p\|$.

The proposed models achieved significant improvements over the 6-Axis IONet in the trajectory where the cumulative error is circular. In the top view, the 6-Axis IONet seems to generate a similar trajectory in the beginning, but the cumulative error occurred owing to the azimuth error in the last part. In the side view, a cumulative error in the z-axis

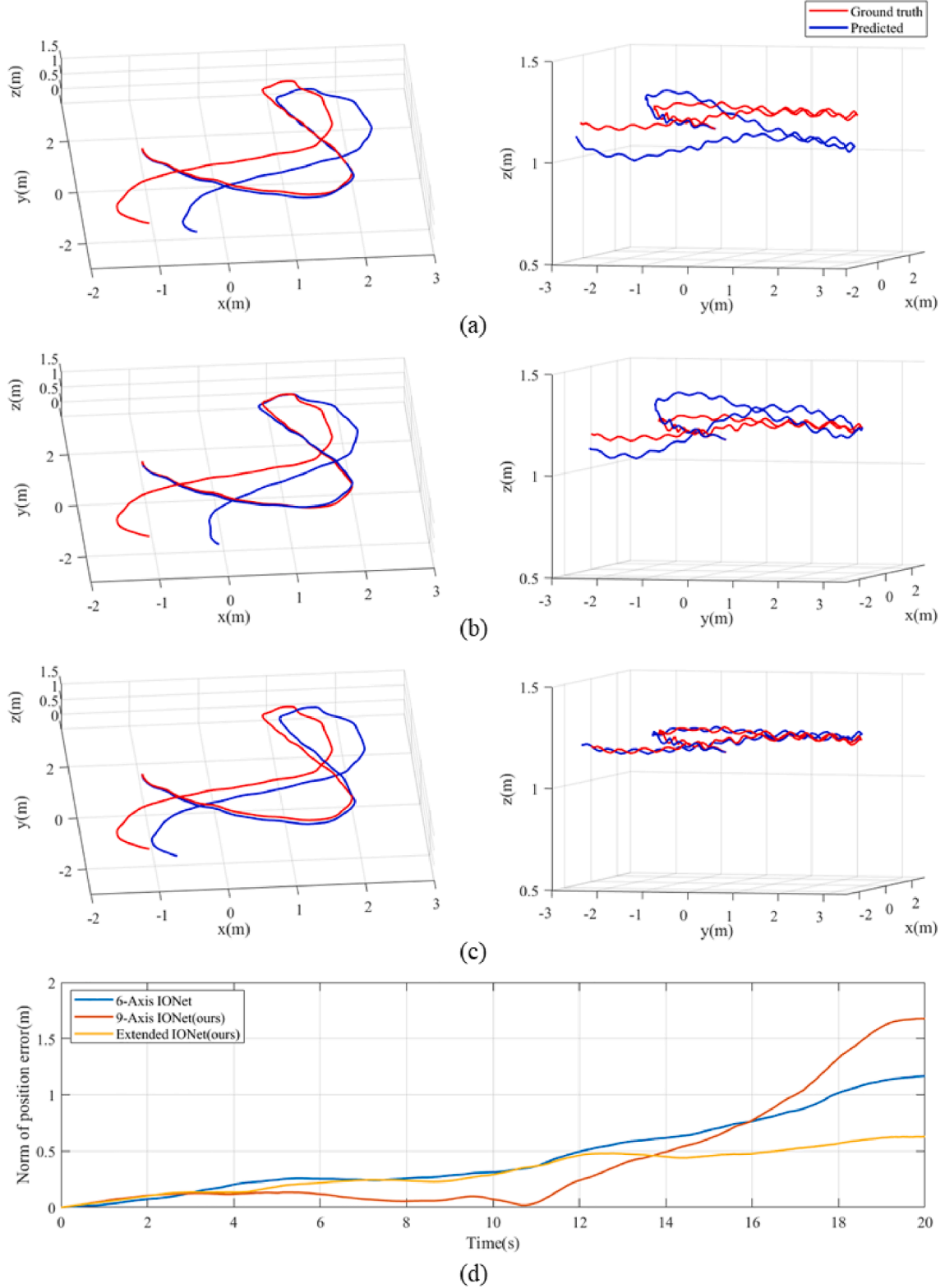


Fig. 7. Top (left column) and side (right column) views of ground truth (blue) and predicted (orange) trajectories for an excerpt of the “handheld/data4/seq3” dataset using (a) 6-Axis IONet, (b) 9-Axis IONet, (c) Extended IONet, and (d) the position error of the trajectories.

direction was generated owing to the error in the inclination angle. The reason for this result is that the 6-Axis IONet only uses linear acceleration and gyro data; thus, the data are not sufficient to obtain the orientation information. The accumulated position errors of the 6-Axis IONet, 9-Axis IONet, and Extended IONet were 1.46, 0.2657, and 0.1991 m, respectively. This experiment showed that the Odometry performance of the proposed models was better than that of the 6-Axis IONet owing to the use of gravitational acceleration and geomagnetism that can affect the z-axis trajectory estimation.

Fig. 7 presents the trajectory-estimation results of the 6-Axis IONet (a), 9-Axis IONet (b), and Extended IONet (c) using the “handheld/data4/seq3” dataset. Fig. 6(d) shows the position error of the trajectories created by the 6-Axis IONet (blue line), 9-Axis IONet (orange line), and Extended IONet (yellow line). The 9-Axis IONet lowered the position error up to 10 s, but the azimuth prediction error increased thereafter. Accordingly, the positional error was the highest from 16 s. In addition, the cumulative error of the Extended IONet, as can be observed from its side view, was significantly improved, compared to those of the 6-Axis IONet and 9-Axis IONet. In addition, the accumulated position errors of the 6-Axis IONet, 9-Axis IONet, and Extended IONet were 1.167, 1.678, and 0.6282 m, respectively. This experiment showed that the generalization performance of Extended IONet was better than those of the 9-Axis IONet and 6-Axis IONet models owing to the direct use of inclination information as a quaternion.

6.2. Quantitative evaluation

To evaluate the reliability of the proposed model, we performed the quantitative evaluation of each model using the RMSE between each model output and the ground truth by inputting 7 test data sequences. Table 2 shows the RMSE of trajectory and orientation predicted by different IONet models. The RMSE of our models are mostly lower than the 6-Axis IONet. Even for the sequence (“handheld/data4/seq1”) where the 6-Axis IONet have the best performance, the difference in the RMSE compared to the models with the 9-Axis IONet and Extended IONet was not too large. The mean RMSEs of the Extended IONet and 9-Axis IONet were 39.8%/37.6% and 33.8%/21.9%, respectively, which were lower than that of the 6-Axis IONet. As a result the 9-Axis IONet can be outperformed that the RMSE could be reduced by adding geomagnetic information and gravitational acceleration containing orientation information as input data. The results for the Extended IONet also confirmed that Pose-TuningNet, which directly converted the gravitational acceleration into the inclination, improved the position estimation performance by correcting the drift occurring in the 9-Axis IONet. The reason why the proposed 9-Axis IONet and Extended IONet show higher performance than the 6-Axis IONet using linear acceleration and gyro is that the acceleration and angular velocity measured by orientation are changed when estimating the change of pose. Therefore, in order to accurately estimate the change of the pose, information on the gravitational acceleration and geomagnetism directly related to the orientation is essential in IONets.

7. Conclusions

In this paper, we proposed the Extended IONet, which is a learning framework connected to the 9-Axis IONet and Pose-TuningNet to improve the performance of the estimated pose changes. By using the gravitational acceleration and geomagnetic data that are excluded in the conventional IONet, we attempted to minimize the errors that occur in the estimation of pose changes. As the proposed Extended IONet uses the inclination information calculated from gravitational acceleration, the trajectory tracking performance can be improved in an environment where cumulative errors can easily occur. To prove the validity of the Extended IONet model, we evaluated the 3D trajectory performance using OxiOD handheld sequences. Our findings showed that the trajectory tracking performance improved by using Pose-TuningNet, which

Table 2

Comparison of RMSE for p/q using the test sequences.

Sequence	6-Axis IONet	9-Axis IONet	Extended IONet
handheld/data1/seq2	0.793/11.892	0.168/3.736	0.132/2.404
handheld/data1/seq5	0.537/10.105	0.219/3.143	0.234/4.089
handheld/data1/seq6	0.411/7.407	0.143/4.088	0.222/3.956
handheld/data3/seq1	0.452/8.196	0.403/4.586	0.501/5.314
handheld/data4/seq1	0.302/20.643	0.413/21.203	0.335/21.928
handheld/data4/seq3	0.565/6.254	0.646/11.801	0.373/1.905
handheld/data5/seq1	0.388/2.197	0.292/3.518	0.285/1.974
Mean	0.493/9.528	0.326/7.439	0.297/5.939

learns information about the orientation using gravitational acceleration data that can be easily obtained from the IMU. Compared with the existing 6-Axis IONet, poor performance was recorded in one out of seven cases; however, the overall RMSE improvement of 39.8% demonstrated the significant effect of our model. When a novel IONet with precise trajectory tracking is developed in the future, the performance can be improved by applying our proposed learning framework.

Deep learning is time consuming, and the intrinsic error and bias of IMUs are likely to be different even among devices of the same model. That is, a trained model that exhibits good performance may still deliver different performances from the same input data, depending on the characteristics of the IMU. Hence, collecting and utilizing a dataset that tracks trajectories for various paths for a new IMU is necessary, which can be costly and time intensive. Therefore, in a future work, we will investigate ways to improve the versatility of IO by designing a model that can be easily used in other devices.

Funding

This work was supported by the Basic Science Research Program through the National Research Foundation of Korea funded by the Ministry of Education [grant number 2019R1I1A3A01063290].

CRedit authorship contribution statement

Won-Yeol Kim: Conceptualization, Methodology, Software, Validation. **Hong-Il Seo:** Writing - original draft, Methodology, Software. **Dong-Hoan Seo:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work presented in this article is based on H.I. Seo's M.S. dissertation (“A Study on real-time 3D Position Estimation based on 9-Axis IMU and Neural Network using Modified IONet,” Graduate School of Korea Maritime and Ocean University).

References

- Alhussein, M., Muhammad, G., & Hossain, M. (2019). EEG Pathology Detection Based on Deep Learning. *IEEE Access*, 7, 27781–27788. <https://doi.org/10.1109/access.2019.2901672>
- Altan, A., & Hacıoğlu, R. (2020). Model predictive control of three-axis gimbal system mounted on UAV for real-time target tracking under external disturbances. *Mechanical Systems and Signal Processing*, 138, 106548. <https://doi.org/10.1016/j.ymssp.2019.106548>
- Altan, A., Karasu, S., & Bekiros, S. (2019). Digital currency forecasting with chaotic meta-heuristic bio-inspired signal processing techniques. *Chaos, Solitons & Fractals*, 126, 325–336. <https://doi.org/10.1016/j.chaos.2019.07.011>
- Belter, D., & Nowicki, M. R. (2019). Optimization-based legged odometry and sensor fusion for legged robot continuous localization. *Robotics and Autonomous Systems*, 111, 110–124.
- Chan, K. Y., Engelke, U., & Abhayasinghe, N. (2017). An edge detection framework conjuncting with IMU data for assisting indoor navigation of visually impaired persons. *Expert Systems with Applications*, 67, 272–284.

- Chen, C., Markham, A., Lu, X., & Trigoni, N. (January 2018). IONet: Learning to cure the curse of drift in inertial odometry. Thirty-Second AAAI conference on Artificial Intelligence.
- Chen, C., Zhao, P., Lu, C. X., Wang, W., Markham, A., & Trigoni, N. (2020). Deep-Learning-Based Pedestrian Inertial Navigation: Methods, Data Set, and On-Device Inference. *IEEE Internet of Things Journal*, 7(5), 4431–4441. <https://doi.org/10.1109/JIoT.648890710.1109/JIoT.2020.2966773>
- Abolfazli Esfahani, M., Wang, H., Wu, K., & Yuan, S. (2020). AbolDeepIO: A novel deep inertial odometry network for autonomous vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 21(5), 1941–1950.
- Forster, C., Zhang, Z., Gassner, M., Werlberger, M., & Scaramuzza, D. (2016). SVO: Semidirect visual odometry for monocular and multicamera systems. *IEEE Transactions on Robotics*, 33(2), 249–265.
- Hannink, J., Kautz, T., Pasluosta, C. F., Barth, J., Schulein, S., Gassmann, K.-G., ... Eskofier, B. M. (2018). Mobile stride length estimation with deep convolutional neural networks. *IEEE Journal of Biomedical and Health Informatics*, 22(2), 354–362.
- Karasu, S., Altan, A., Bekiros, S., & Ahmad, W. (2020). A new forecasting model with wrapper-based feature selection approach using multi-objective optimization technique for chaotic crude oil time series. *Energy*, 212, 118750. <https://doi.org/10.1016/j.energy.2020.118750>
- Kendall, A., Gal, Y., & Cipolla, R. (2018). Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. *Proceedings of the of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Kone, Y., Zhu, N., Renaudin, V., & Ortiz, M. (2020). Machine Learning-Based Zero-Velocity Detection for Inertial Pedestrian Navigation. *IEEE Sensors Journal*, 20(20), 12343–12353.
- Li, D., Zhao, D., Zhang, Q., & Chen, Y. (2019). Reinforcement Learning and Deep Learning Based Lateral Control for Autonomous Driving. *IEEE Computational Intelligence Magazine*, 14(2), 83–98. <https://doi.org/10.1109/mci.2019.2901089>
- Li, M., & Mourikis, A. I. (May 2012). Improving the Accuracy of EKF-based Visual-Inertial Odometry. In *Proceedings of the IEEE International Conference on Robotics and Automation* (pp. 828–835).
- Lu, C., Uchiyama, H., Thomas, D., Shimada, A., & Taniguchi, R.-I. (2019). Indoor positioning system based on chest-mounted IMU. *Sensors*, 19(2), 420. <https://doi.org/10.3390/s19020420>
- Mannini, A., & Sabatini, A. M. (February 2010). Machine learning methods for classifying human physical activity from on-body accelerometers. *Sensors*, 10(2), 1154–1175. <https://doi.org/10.3390/s100201154>
- Mannini, A., & Sabatini, A. M. (October 2014). Walking speed estimation using foot-mounted inertial sensors: Comparing machine learning and strap-down integration methods. *Medical Engineering and Physics*, 36(10), 1312–1321. <https://doi.org/10.1016/j.medengphy.2014.07.022>
- Marchand, E., Uchiyama, H., & Spindler, F. (2015). Pose estimation for augmented reality: A hands-on survey. *IEEE transactions on visualization and computer graphics*, 22(12), 2633–2651.
- Nilsson, J. O., Gupta, A. K., & Händel, P. (2014). Foot-mounted inertial navigation made easy. *IEEE International conference on Indoor Positioning and Indoor Navigation (IPIN)*.
- Nilsson, J.-O., Zachariah, D., Skog, I., & Händel, P. (2013). Cooperative localization by dual foot-mounted inertial sensors and inter-agent ranging. *EURASIP Journal on Advances in Signal Processing*, 2013(1). <https://doi.org/10.1186/1687-6180-2013-164>
- Parate, A., Chiu, M. C., Chadowitz, C., Ganesan, D., & Kalogerakis, E. (2014). In RisQ: Recognizing smoking gestures with inertial sensors on a wristband (pp. 149–161). and Services (MobiSys). <https://doi.org/10.1145/2594368.2594379>.
- Parcollet, T., Morchid, M., & Linares, G. (April 2020). A survey of quaternion neural networks. *Artificial Intelligence Review*, 53(4), 2957–2982. <https://doi.org/10.1007/s10462-019-09752-1>
- Savage, P. G. (1998). Strapdown inertial navigation integration algorithm design part 1: Attitude algorithms. *Journal of guidance, control, and dynamics*, 21(1), 19–28.
- Scaramuzza, D., & Fraundorfer, F. (2011). Visual odometry [tutorial]. *IEEE robotics & automation magazine*, 18(4), 80–92.
- Seong, J. H., Lee, S.-H., Yoon, K.-K., & Seo, D.-H. (2019). Ellipse coefficient map-based geomagnetic fingerprint considering azimuth angles. *Symmetry*, 11(5, May). <https://doi.org/10.3390/sym11050708>.
- Silva do Monte Lima, J. P., Uchiyama, H., & Taniguchi, R.-I. (2019). End-to-end learning framework for IMU-based 6-dof odometry. *Sensors*, 19(17), 3777. <https://doi.org/10.3390/s19173777>
- Titterton, D., Weston, J. L., & Weston, J. (2004). Strapdown Inertial Navigation Technology. IET, 17.
- Valtazanos, A., Arvind, D. K., & Ramamoorthy, S. (2013). Using wearable inertial sensors for posture and position tracking in unconstrained environments through learned translation manifolds. *IEEE Academic Medicine/IEEE International Conference on Information Processing in Sensor Networks, IPSN*, 241–252.
- Wagstaff, B., Peretroukhin, V., & Kelly, J. (2020). Robust Data-Driven Zero-Velocity Detection for Foot-Mounted Inertial Navigation. *IEEE Sensors Journal*, 20(2), 957–967.
- Wang, Y., Cang, S., & Yu, H. (2019). A survey on wearable sensor modality centred human activity recognition in health care. *Expert Systems with Applications*, 137, 167–190.
- Wu, Y., Hu, X., Hu, D., Li, T., & Lian, J. (2005). Strapdown inertial navigation system algorithms based on dual quaternions. *IEEE Transaction on Aerospace and Electronic Systems*, 41(1), 110–232.
- Yan, H., Shan, Q., & Furukawa, Y. (2018). RIDI: Robust IMU double integration. Lecture Notes in Computer Science. In *Proceedings of the of the European Conference on Computer Vision (ECCV)* (pp. 641–656). https://doi.org/10.1007/978-3-030-01261-8_38
- Yuwono, M., Su, S. W., Guo, Y., Moulton, B. D., & Nguyen, H. T. (January 2014). Unsupervised nonparametric method for gait analysis using a waist-worn inertial sensor. *Applied Soft Computing*, 14(A), 72–80. <https://doi.org/10.1016/j.asoc.2013.07.027>.
- Zihajezadeh, S., Loh, D., Lee, T. J., Hoskinson, R., & Park, E. J. (2015). A cascaded Kalman filter-based GPS/MEMS-IMU integration for sports applications. *Measurement*, 73, 200–210.